

Problem 5

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Problem 1 Find the derivative of the following functions:

$$(a) F(x) = \int_{\sqrt{x}}^1 \frac{t^2}{2+3t^4} dt$$

Explanation. First notice that

$$F(x) = \int_{\sqrt{x}}^1 \frac{t^2}{2+3t^4} dt = - \int_1^{\sqrt{x}} \frac{t^2}{2+3t^4} dt.$$

So we can apply the First Fundamental Theorem of Calculus, along with the chain rule, to compute:

$$F'(x) = \frac{d}{dx} \left(- \int_1^{\sqrt{x}} \frac{t^2}{2+3t^4} dt \right)$$

$$\begin{aligned} \text{we introduce the variable: } u = \sqrt{x} \implies F'(x) &= \frac{dF}{du} \frac{du}{dx} \\ &= - \frac{u^2}{2+3(u)^4} \cdot \frac{d}{dx}(\sqrt{x}) \\ &= - \frac{u^2}{2+3u^4} \cdot \frac{1}{2\sqrt{x}} \end{aligned}$$

$$\begin{aligned} \text{substituting } u = \sqrt{x} \implies F'(x) &= - \frac{x}{2+3x^2} \cdot \frac{1}{2\sqrt{x}} \\ &= - \frac{\sqrt{x}}{2(2+3x^2)} \end{aligned}$$

$$(b) G(x) = \int_x^{x^3} \sin(7t^2) dt$$

Explanation. First notice that

$$\begin{aligned} G(x) &= \int_x^{x^3} \sin(7t^2) dt \\ &= \int_x^0 \sin(7t^2) dt + \int_0^{x^3} \sin(7t^2) dt \\ &= - \int_0^x \sin(7t^2) dt + \int_0^{x^3} \sin(7t^2) dt \end{aligned}$$

It is worth pointing out that breaking up the integral above at 0 was arbitrary. Since $\sin(7t^2)$ is continuous over $(-\infty, \infty)$, we could have chosen any real number in place of 0.

We can now apply the First Fundamental Theorem of Calculus, along with the chain rule, to compute:

$$\begin{aligned} G'(x) &= \frac{d}{dx} \left(- \int_0^x \sin(7t^2) dt + \int_0^{x^3} \sin(7t^2) dt \right) \\ &= \frac{d}{dx} \left(- \int_0^x \sin(7t^2) dt \right) + \frac{d}{dx} \left(\int_0^{x^3} \sin(7t^2) dt \right) \end{aligned}$$

The left integral requires no chain rule on the FFTOC. $\frac{d}{dx} \left(- \int_0^x \sin(7t^2) dt \right) = -\sin(7x^2)$

The right integral requires chain rule. We'll substitute $u = x^3$.

$$\begin{aligned} \frac{d}{dx} \left(\int_0^{x^3} \sin(7t^2) dt \right) &= \frac{d}{du} \left(\int_0^u \sin(7t^2) dt \right) \frac{du}{dx} \\ &= \sin(7u^2) \cdot \frac{d}{dx}(x^3) \\ &= \sin(7u^2) \cdot 3x^2 \end{aligned}$$

$$\text{substituting } u = x^3 \implies \frac{d}{dx} \left(\int_0^{x^3} \sin(7t^2) dt \right) = 3x^2 \sin(7x^6)$$

Combining we have:

$$G'(x) = 3x^2 \sin(7x^6) - \sin(7x^2).$$

An alternative approach using SFTOC: Call $h(t) = \sin(7t^2)$ and let H be an antiderivative of h . By SFTOC, $\int_x^{x^3} \sin(7t^2) dt = \left[H(t) \right]_{t=x}^{x^3}$.

Then:

$$\begin{aligned}
 \frac{d}{dx} \left(\int_0^{x^3} \sin(7t^2) dt \right) &= \frac{d}{dx} \left(\left[H(t) \right]_{t=x}^{x^3} \right) \\
 &= \frac{d}{dx} (H(x^3) - H(x)) \\
 &= H'(x^3) \cdot 3x^2 - H'(x) \\
 &= h(x^3) \cdot 3x^2 - h(x) \\
 &= \sin(7x^6) \cdot 3x^2 - \sin(7x^2)
 \end{aligned}$$